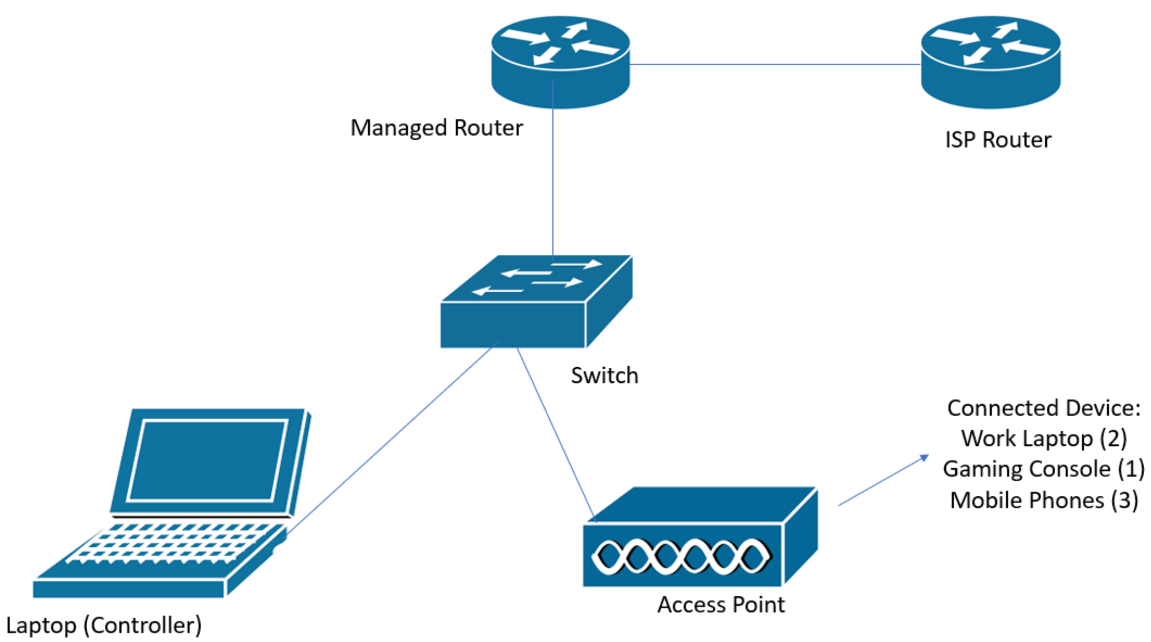
VLAN-Segmented Home Network Lab

Description
The project consists of a small VLAN-segmented network infrastructure implemented using Omada-managed devices to support separate administrative, home, and work network environments within a single physical location.

Network Topology



Devices Used

Device	Brand	Model	Type
Router	Skyworth	GN630V	Infrastructure Device
Router	Omada by TP-Link	ER605	Infrastructure Device
Switch	TP-Link	TL-SG108E	Infrastructure Device
Omada by TP-Link	Access Point	EAP610 AX1800	Infrastructure Device
Laptop	MSI	Modern 15 B5M	End Device
Ethernet Cables	NA	CAT5e	Network Medium
Omada Controller Platform	Omada by TP-Link	NA	Controller Software

Networks Established

VLAN ID	Name	Access
1	Default	Wired and Wireless
10	RT (For Management)	Wired and Wireless
20	Home	Wired and Wireless
30	Work	Wired and Wireless

Network Descriptions

Default Network (VLAN 1) - The Default VLAN is the preconfigured network segment provided by the ISP router prior to the implementation of VLAN segmentation within the environment.

RT Management Network (VLAN 10) - The RT VLAN was implemented as a dedicated management network intended for controller management, infrastructure administration, and device configuration. Administrative devices and Omada-managed infrastructure components reside within this VLAN to centralize management traffic and separate administrative access from user networks. The management VLAN was intentionally labeled with a non-descriptive identifier to reduce unnecessary exposure of its administrative purpose within the environment.

Home Network (VLAN 20) - The Home VLAN was configured to provide network connectivity for home devices and IoT (Internet of Things) systems separated from administrative and work-related traffic.

Work Network (VLAN 30) - The Work VLAN was configured to provide dedicated connectivity for work-related systems and remote work devices while maintaining segmentation from home and management networks.

Troubleshooting Steps Encountered during Network Implementation

I. Devices Unable to Connect to the Omada Controller platform.

Problem:
The Omada-managed router and access point failed to connect and appear for adoption within the Omada Controller platform

Overview:
During the initial deployment stage, two separate VLAN-based networks were configured in addition to the existing Default Network, resulting in the following active network segments:

- Default
- Home
- Work

At the time of deployment, the Omada Controller platform host (laptop) was connected to Switch Port 1 assigned to the Home network, while the Omada router and access point were connected to Ports 2 and 4 assigned to the Default network.

Switch Port Network Assignment:

Port	Network	Device Connected
1	Home	Laptop
2	Default	Router
3	Default	Access Point
4	Home	None
5	Home	None
6	Work	None
7	Work	None
8	Work	None

Cause:
The Omada Controller platform host and the Omada-managed devices were assigned to separate network segments, preventing the management traffic required for device discovery and controller adoption.
Since the controller host resided on the Home network while the router and access point resided on the Default network, Layer 2 communication between the devices was unavailable. At this stage of deployment, no inter-network routing path had been configured to allow management connectivity between the two network segments.
As a result, the Omada Controller platform was unable to discover and adopt the managed router and access point.

Solution:
Switch Port 1 was reassigned from the Home network to the Default network to place the Omada Controller platform host and managed devices within the same network segment.
After reassigning the switch port, the router and access point were restarted to re-establish controller discovery and management communication. The devices were then successfully detected and adopted within the Omada Controller platform.

Validation:

- The Omada-managed router and access point successfully appeared within the Omada Controller platform for adoption.
- Controller communication between the Omada Controller host and managed devices was successfully established.
- Infrastructure devices remained reachable after reassignment to the same network segment.
- Device adoption and centralized management functions operated successfully after the switch port reassignment.

II. Omada-Managed Devices Disconnected from the Omada Controller Platform

Problem:
The Omada-managed router and access point lost connectivity with the Omada Controller platform after the implementation of a dedicated management network.

Overview:
To improve network segmentation and administrative security, a dedicated management VLAN named RT with VLAN ID 10 was created for controller management and infrastructure administration.
At this stage of the configuration, the switch port assignments were configured as follows:

Port	Network	VLAN ID	Device Connected
1	RT	10	Laptop
2	Default	1	Router
3	Default	1	Access Point
4	Home	20	None
5	Home	20	None
6	Work	30	None
7	Work	30	None
8	Work	30	None

Cause:
The Omada Controller platform host and the Omada-managed devices were once again assigned to separate VLANs. The laptop hosting the Omada Controller platform resided on VLAN 10, while the router and access point remained connected to the Default VLAN.

Since no management routing path was configured between the two VLANs, controller communication and device management traffic could not traverse the network boundary, resulting in loss of controller connectivity.

Solution:
The switch port assignments were reconfigured to place all infrastructure management devices within the RT management VLAN.

The Default VLAN was removed from the infrastructure device port assignments, leaving only the following operational VLANs within the environment:

- RT (Management)
- Home
- Work

Updated TL-SG108E Switch Port Assignment

Port	Network	VLAN ID	Device Connected
1	RT	10	Laptop
2	RT	10	Router
3	RT	10	Access Point
4	Home	20	None
5	Home	20	None
6	Work	30	None
7	Work	30	None
8	Work	30	None

After reassigning the ports to the management VLAN, connectivity between the Omada Controller platform and managed devices was successfully restored.

Validation:

- Connectivity between the Omada Controller and managed infrastructure devices was successfully restored after migration to the RT management VLAN.
- The Omada Controller maintained stable communication with the router and access point within VLAN 10.
- Administrative traffic was successfully separated from Home and Work VLAN traffic.
- Infrastructure management functionality remained operational after removal of the Default VLAN from the management device ports.

III. Access Point Unable to Provide Connectivity for Multiple Networks

Problem:
The Omada access point was only able to provide wireless connectivity for the RT management network and was unable to distribute connectivity for the Home and Work VLANs.

Overview:
At this stage of deployment, the switch ports connected to the router and access point were configured as untagged access ports assigned to VLAN 10.

Cause:
Ports 2 and 3 were configured as untagged access ports for VLAN 10. Because access ports only carry traffic for a single VLAN, the router and access point were unable to transmit VLAN-tagged traffic for the Home and Work networks.

As a result, only RT VLAN traffic was forwarded through the infrastructure links connected to the router and access point.

Switch Ports Initial Configuration:

Port	**Network**	**VLAN ID**	**Tag/Untag**
1	RT	10	Untag - Access
2	RT	10	Untag- Access
3	RT	10	Untag- Access
4	Home	20	Untag- Access
5	Home	20	Untag- Access
6	Work	30	Untag- Access
7	Work	30	Untag- Access
8	Work	30	Untag- Access

Solution:
Ports 2 and 3 were reconfigured from untagged access ports to tagged trunk ports to allow VLAN-tagged traffic for multiple network segments to traverse the infrastructure uplinks.

This configuration enabled the router and access point to carry and distribute traffic for VLANs 10, 20, and 30 while maintaining network segmentation between the respective VLANs.

Switch Ports Updated Configuration

Port	**Network**	**VLAN ID**	**Tag/Untag**
1	RT	10	Untag – Access
2	RT	10	Tag- Trunk
3	RT	10	Tag- Trunk
4	Home	20	Untag- Access
5	Home	20	Untag- Access
6	Work	30	Untag- Access
7	Work	30	Untag- Access
8	Work	30	Untag- Access

Ports 2 and 3 now function as trunk links between the switch, router, and access point, enabling multiple VLANs to traverse a single physical connection and allowing the access point to distribute wireless connectivity across all configured network segments.

Validation:

- Wireless clients successfully connected to the RT, Home, and Work wireless networks.
- VLAN-tagged traffic for VLANs 10, 20, and 30 successfully traversed the trunk links connected to the router and access point.
- The access point successfully distributed wireless connectivity across multiple VLANs through a single uplink connection.
- Network segmentation between VLANs remained operational while maintaining wireless connectivity for all configured network environments.

Lesson Learned:

- Importance of proper VLAN planning
- Difference between tagged and untagged ports
- Importance of management VLAN separation
- Understanding Layer 2 VLAN boundaries
- Trunk port configuration for multi-VLAN environments
- Importance of topology and infrastructure documentation

-End of Documentation-